Model Explainability

What is Model Explainability?

- Understanding how and why a model makes predictions
- Critical for trust, validation, debugging, and regulatory compliance
- Especially important for black-box models

Types of Explainability

- **Global Explainability**: Overall model behavior
- **Local Explainability**: Single prediction explanation

Approaches → Next slides

Impurity-Based Feature Importance

- Built into tree-based models
- Measures reduction in impurity during training
- Global only
- Biased toward high-cardinality features

Reference

Permutation Feature Importance

- Shuffles feature values and measures performance drop
- Model-agnostic
- Can generate unrealistic feature combinations
- Works for global explanations

Reference

Leave One Feature Out (LOFO)

- Removes a feature and retrains the model
- Measures performance degradation
- Model-agnostic
- Computationally expensive
- Sensitive to correlated features

Reference

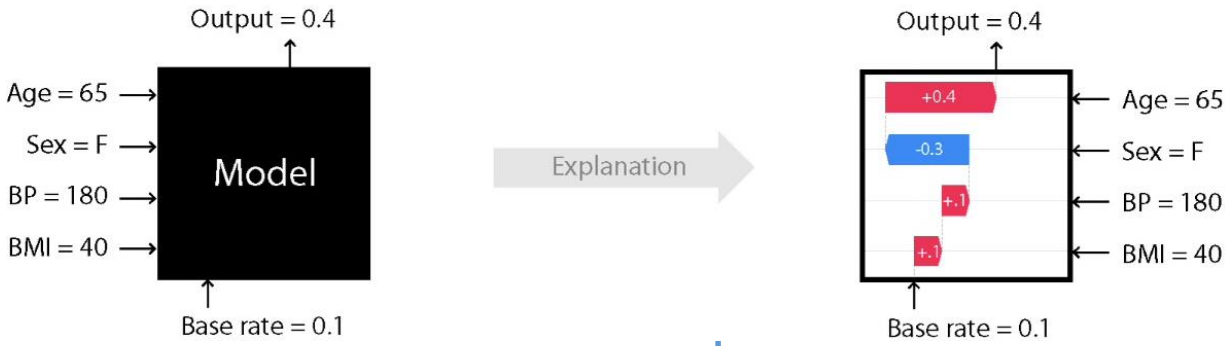
LIME (Local Interpretable Model-Agnostic Explanations)

- Trains a local surrogate model
- Explains individual predictions
- Works with tabular, text, and image data
- Dependent on surrogate model fidelity

Reference

SHAP (Shapley Additive Explanations)

- Based on game theory
- Provides both global and local explanations
- Captures feature interactions
- Efficient for tree-based models (TreeExplainer)
- Considered robust and flexible



SHAP (Shapley Additive Explanations)

A machine learning model's prediction, f(x), can be represented as the sum of its computed SHAP values, plus a fixed base value, such that:

$$f(x) = base\ value + sum(SHAP\ values)$$

- For regression models, the base value is equal to the mean of the target variable.
- For classification models, the base value is equal to the prevalence of the positive class.
- SHAP values provide valuable insights into how input variables influence the machine learning model's predictions, both for individual cases and for the overall population.

SHAP (Shapley Additive Explanations)

Local interpretability: explaining individual predictions

Explaining predictions for individual data instances is referred to as local interpretability. SHAP explains how individual predictions are derived by summing the contributions of each model input variable. This is a highly intuitive approach that produces simple but informative outputs.

Global interpretability: understanding drivers of predictions across the population

Global interpretation methods aim to describe a machine learning model's overall behavior across all possible input values. SHAP accomplishes this by aggregating the SHAP values from individual instances over the entire dataset.

SHAP (Shapley Additive Explanations)

SETTING-UP

SHAP can be installed from either PyPI or conda-forge:

```
pip install shap
or
conda install -c conda-forge shap
```

How to use?

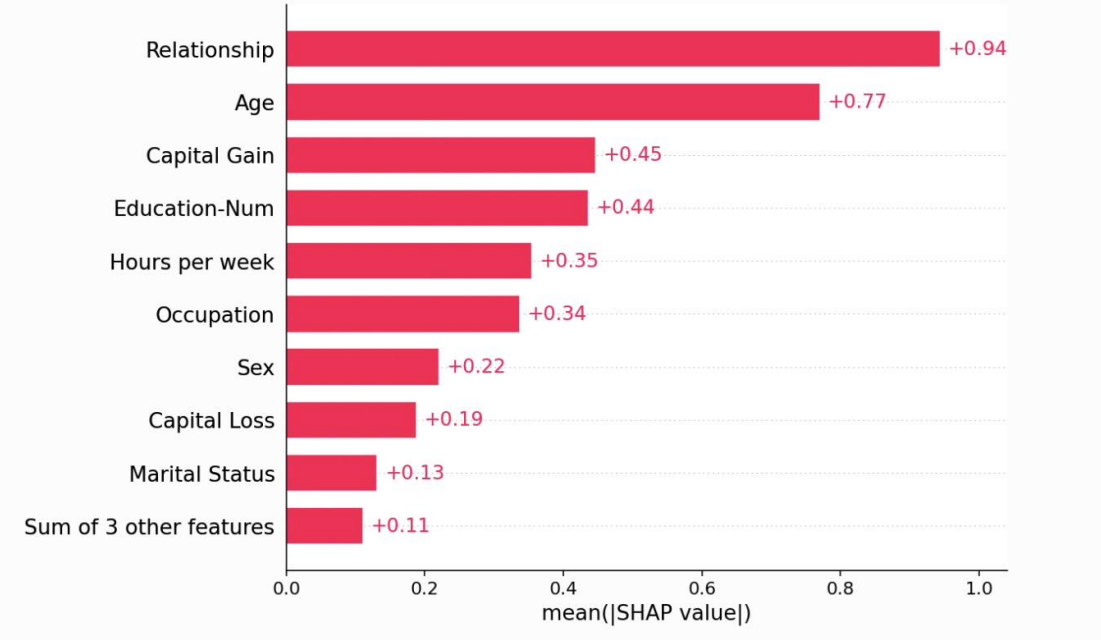
```
import shap
```

[Reference]

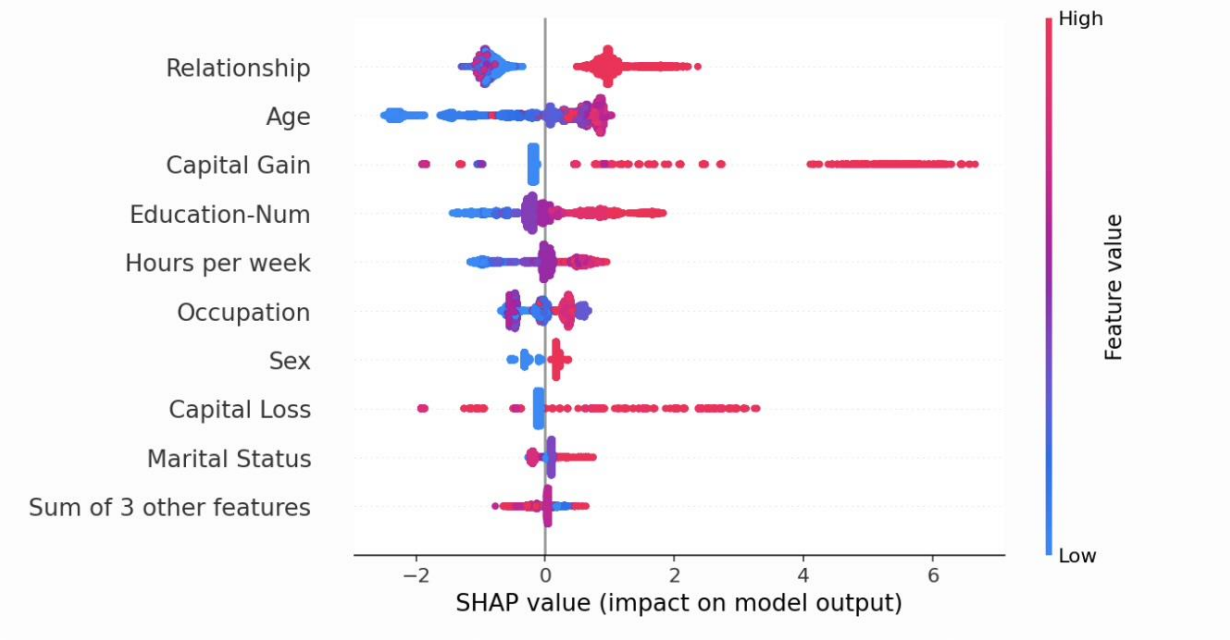
SHAP (Shapley Additive Explanations)

- Different ways to plot it

```
shap.plots.bar(shap_values)
```



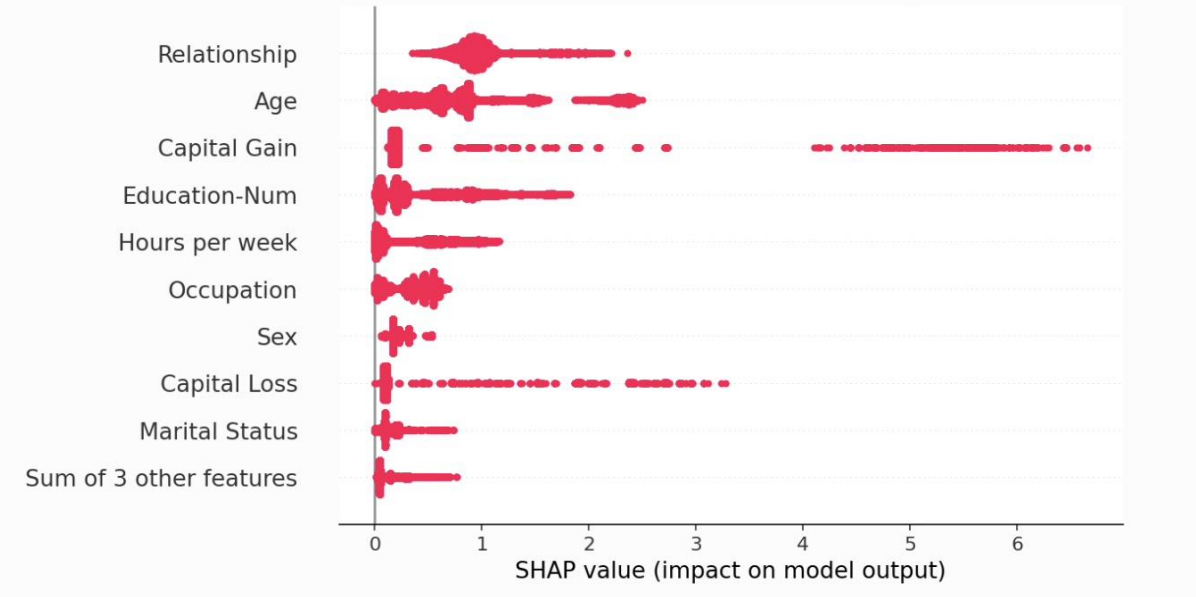

```
shap.plots.beeswarm(shap_values)
```



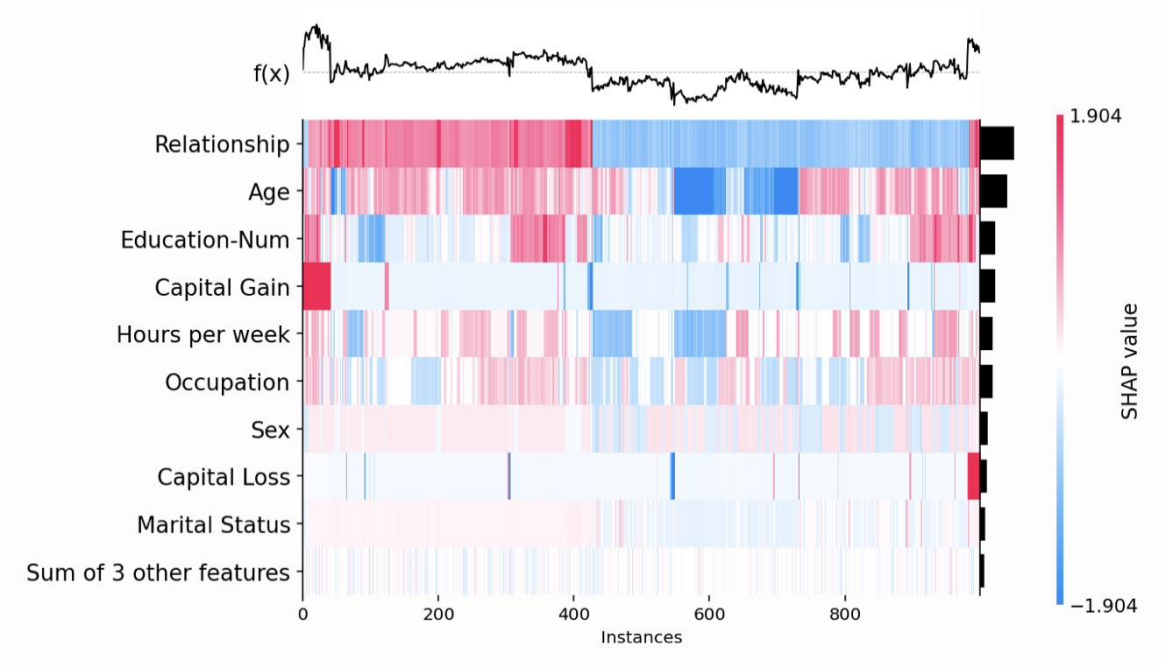
SHAP (Shapley Additive Explanations)

- Different ways to plot it

```
shap.plots.beeswarm(shap_values.abs, color="shap_red")
```




```
shap.plots.heatmap(shap_values[:1000])
```



Comparison Considerations

- Explanation type needed (global vs local)
- Model compatibility
- Handling correlated features
- Computational cost
- Data perturbation realism